mGTE: Generalized Long-Context Text Representation and Reranking Models for Multilingual Text Retrieval

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Abstract

We present systematic efforts in building long-context multilingual text representation model (TRM) and reranker from scratch for text retrieval. We first introduce a text encoder (base size) enhanced with RoPE and unpadding, pre-trained in a native 8192-token context (longer than 512 of previous multilingual encoders). Then we construct a hybrid TRM and a cross-encoder reranker by contrastive learning. Evaluations show that our text encoder outperforms the same-sized previous state-of-the-art XLM-R. Meanwhile, our TRM and reranker match the performance of large-sized state-of-the-art BGE-M3 models and achieve better results on long-context retrieval benchmarks. Further analysis demonstrate that our proposed models exhibit higher efficiency during both training and inference. We believe their efficiency and effectiveness could benefit various researches and industrial applications.[1]

1 Introduction

Text retrieval aims to find relevant passages or documents from a large corpus given a query (Manning, 2008). It is often implemented as a multi-stage process, consisting of two main components: a *retriever* and a *reranker* (Gao et al., 2021a; Zhang et al., 2022; Zhao et al., 2024). The retriever identifies a set of candidate documents that are potentially relevant to the query based on the similarity between their sparse (lexical term weights) or/and dense representations from a text representation model (TRM). While the reranker reorders these retrieved candidates to refine the results based on the relevance score generated by a more precise yet computationally demanding model that processes both the query and a candidate document together.

Recent advances in large language models (LLMs) and retrieval augmented generation (RAG) (Gao et al., 2023) systems have led to an unprecedented surge in demand for versatile, plug-and-play TRMs and rerankers. These new applications heavily involve processing long and multilingual texts, which could not be addressed by conventional encoder-based models and urgently require upgraded ones. To this end, some resort to enhancing existing multilingual encoders, *e.g.,* XLM-R (Conneau et al., 2020), with extended context window up to 8192 (Chen et al., 2024). Others turn to use multilingual LLMs which already have the required capabilities (Zhang et al., 2023a), but their models might be computationally expensive for self-hosted search services.

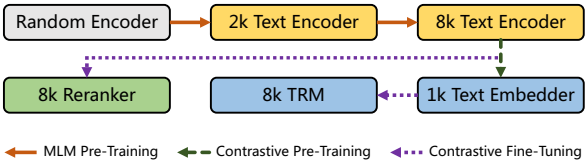

Figure 1: Training pipeline. We first build an 8k long-context multilingual encoder. Then based on it, we train text representation and reranking models for retrieval.

In the English community, it has been proven that training long-context encoders from scratch is promising for text retrieval (Günther et al., 2023; Nussbaum et al., 2024). In this work, we continue this journey, presenting systematic efforts in building the long-context multilingual text encoder, TRM, and reranker. We suggest a holistic pipeline (Figure 1) as well as several techniques in modeling and training for multilingual long-context retrieval.

Concretely, we first introduce a text encoder enhanced with Rotary Position Embedding (RoPE, Su et al., 2024) and unpadding (Portes et al., 2023), pre-trained by masked language modeling (MLM) (Devlin et al., 2019) via a two-stage curriculum for the native 8,192 tokens context. Based on our encoder, we propose a hybrid TRM capable of generating both elastic dense (Kusupati et al., 2022)

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[1]Models are released at https://hf.co/Alibaba-NLP/gte-multilingual-base.

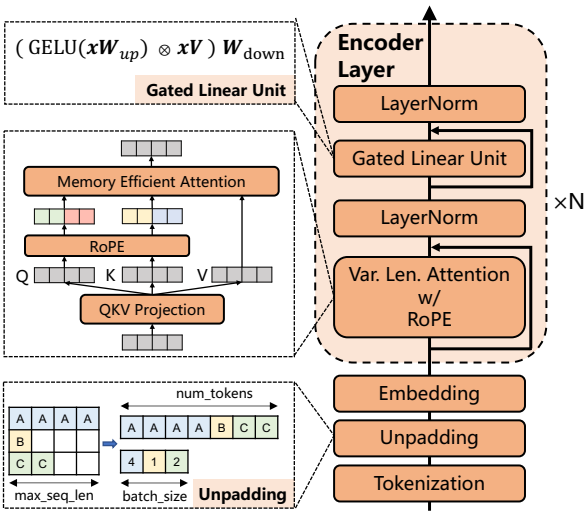

Figure 2: Our text encoder architecture.

and sparse vectors for efficient first-stage retrieval, as well as a cross-encoder reranker. We construct them via the contrastive learning objective (Wang et al., 2022; Li et al., 2023) with large-scale meticulously curated datasets, providing robust off-the-shelf retrieval models.

We conduct extensive experiments to verify our method. For the text encoder, we evaluate on two natural language understanding (NLU) benchmarks, *i.e.,* XTREME-R (Ruder et al., 2021) and GLUE (Wang et al., 2018), and show that our encoder outperforms the same-sized previous state-of-the-art XLM-R. For the TRM and reranker, we evaluate on multiple retrieval benchmarks with multilingual and long-context settings, *e.g.,* MIRACL (Zhang et al., 2023b) and MLDR (Chen et al., 2024), where our models match the performance of state-of-the-art BGE-M3 (Chen et al., 2024) and achieve better long-context performance by a smaller size. We open-source our models and code to facilitate further research and applications.

2 Method

2.1 Text Encoder

To construct powerful long-context multilingual text encoder models, we implement several enhancements to BERT (Devlin et al., 2019) architecture and train it from scratch using the vocabulary of XLM-R[2] (Conneau et al., 2020) series.

Specifically, we replace the absolute positional embeddings with RoPE (Su et al., 2024), and upgrade the feedforward network (FFN) to gated linear unit (GLU) (Shazeer, 2020). To ensure compatibility with libraries like `FlashAttention` (Dao, 2023), we remove the dropout applied to attention scores. In addition, we pad the token embedding size to be a multiple of 64, which could speedup the model throughput (Portes et al., 2023).

Unpadding Mode Inspired by Portes et al. (2023), we unpad the input batch to reduce redundant computations associated with padding tokens (Figure 2). We use `xFormers` (Lefaudeux et al., 2022) to implement the variable length attention. It dispatch the attention forward and backward to different kernels[3] based on the numerical precision, attention head size and device type. We unpad the MLM labels as well to reduce the computation cost of predicting non-masked tokens.

Data We assemble our multilingual pre-training data from a combination of the following sources: C4 (Raffel et al., 2020), Skypile (Wei et al., 2023) (2021-2023 subsets), mC4 (Xue et al., 2021), CulturaX (Nguyen et al., 2024), Wikipedia (Foundation) and books (proprietary). We filter them and curate a dataset covering 75 Languages. Appendix Table 7 presents the statistics of our dataset.

Training Curriculum We pre-train the model via masked language modeling (MLM) (Devlin et al., 2019)[4]. The MLM probability is set to 30% (Portes et al., 2023). Following Conneau and Lample (2019) and Conneau et al. (2020), the data from different languages is sampled by a multinomial distribution with probabilities $\{q_i\}_{i=1\ldots N}$, where

$$q_i = \frac{p_i^\alpha}{\sum_{j=1}^{N} p_j^\alpha} \quad \text{with} \quad p_i = \frac{n_i}{\sum_{j=1}^{N} n_j}, \qquad (1)$$

and n_i is the number of texts in language i. We set $\alpha = 0.5$. This sampling strategy could increase texts from low-resource languages. To train the native 8192-context model more efficiently, we adopt a phased training curriculum (Xiong et al., 2024):

- MLM-2048: we chunk the input into 2048 tokens and set RoPE base to $10,000$.
- MLM-8192: we chunk the input into 8192 tokens and set RoPE base to $160,000$.

Through this method, we could train the model with a large context length in limited resources [5].

[2] `https://hf.co/FacebookAI/xlm-roberta-base`

[3] We adopt the memory-efficient attention (Rabe and Staats, 2021) in this work.

[4] We remove the next sentence prediction objective of BERT following (Liu et al., 2019).

[5] In our early experiments of English models, we investigated continue training by RetroMAE (Xiao et al., 2022) after MLM-8192. However, we did not observe any improvement.

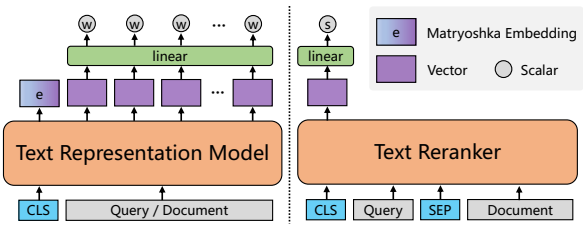

Figure 3: Our TRM and reranker.

Training Setup Following Portes et al. (2023), we use the learning rate decoupled[6] AdamW (Loshchilov and Hutter, 2018) with weight decay $1e-5$. We disable gradient clipping (set to 0) (Liu et al., 2019). All models are trained on A100 GPU servers by BF16 PyTorch native automatic mixed precision via `transformers` (Wolf et al., 2020). We list the detailed hyper-parameters of each training stage in Appendix A.2 and Table 8. We denote the resulting models as `mGTE-MLM-2048/8192`.

2.2 Text Representation Model

Based on our encoder, we construct the TRM for the first-stage text retrieval in two steps: contrastive pre-training and fine-tuning (Wang et al., 2022; Li et al., 2023). Both steps share the same InfoNCE (Oord et al., 2018) learning objective:

$$\mathcal{L} = -\log \frac{\exp(s(q, d^+)/\tau)}{\sum_{i=1}^{N} \exp(s(q, d^i)/\tau)}, \qquad (2)$$

where τ, q, and d denote the temperature parameter, query and document. The positive d^+ is the relevant document to q, and other irrelevant documents are negatives. These negatives can be either hard-negatives or in-batch negatives (documents of other instances in the same batch). $s(q, d)$ is the relevance score of q and d, measured by the dot product or cosine similarity between their respective representations.

Contrastive Pre-Training We take the encoder output hidden state of the [CLS] token as the dense representation (*i.e.,* embedding) and compute the relevance score by cosine similarity. Our pre-training data (Appendix Table 9) comprise naturally occurring text pairs (*e.g.,* question-answer pairs from Quora and StackExchange, title-content pairs of CommonCrawl), translation pairs (Team et al., 2024), and crosslingual instruction tuning data (Muennighoff et al., 2023b). We train the model with a batch size of $16,384$ and a learning rate of $5e-4$ for $240k$ steps. Each batch is sampled from a single data source by the same distribution of Eq.1. The queries (*resp.* documents) are truncated to the max tokens of 512 (*resp.* 1024). We reverse scale the RoPE base from $160,000$ to $20,000$ to fit the 1024 context length and acquire the long-context retrieval ability (denotes revNTK, ablation in §3.4). We set τ of InfoNCE to 0.01 and only use in-batch negatives. More details refer to Appendix B.3. We denote this contrastive pre-trained model as `mGTE-CPT`, which is actually an unsupervised embedding model.

Matryoshka Embedding Many of recently released models and APIs offer elastic embeddings by Matryoshka representation learning (MRL) (Kusupati et al., 2022), providing competitive sub-vectors of embeddings to save index storage and speedup search. Let $\boldsymbol{e} \in \mathbb{R}^H$ denotes an embedding and $\boldsymbol{e}_{:d}$ is the sliced sub-vector from dimension 0 to $d < H$. MRL[7] optimizes the weighted sum of multiple losses from different d dimensional sub-vectors, *i.e.,* compute InfoNCE by $s_d(\boldsymbol{e}^q_{:d}, \boldsymbol{e}^d_{:d})$. We add this objective to our TRM fine-tuning stage.

Sparse Representation Chen et al. (2024) show that neural sparse representations (term/token weights predicted by TRM) could greatly improve the long-context retrieval performance. We follow this design, computing the term weight w_t of each token of the input by $w_t = \text{ReLU}(\boldsymbol{W}\boldsymbol{h}_t)$, where $\boldsymbol{h}_t$ is the encoder hidden state of token t with dimension size H and $\boldsymbol{W} \in \mathbb{R}^{H\times 1}$ is randomly initialized. If a token appears multiple times in the text, we keep the max weight. The relevance score is computed by the joint importance of the co-occurring terms (denoted as $q \cap d$) within the query and document pair: $s_{\text{sparse}}(q, d) = \sum_{t \in q \cap d}(w^q_t \cdot w^d_t)$. This is then used to derive the InfoNCE loss for training.

Contrastive Fine-Tuning Now we construct the TRM by multi-task learning of matryoshka embedding and sparse representation:

$$\mathcal{L}_{\text{TRM}} = \lambda \mathcal{L}_{\text{sparse}} + \sum_{d \in D} w_d \mathcal{L}_{:d}\,, \qquad (3)$$

where $D = \{32k \mid k \in \mathbb{N}, k \geq 1, 32k \leq H\}$ is MRL dimension set, w_d is the weight of dimension d, and λ is the weight of sparse representation loss.

[6]However, Xie et al. (2023b) state that the decoupled weight decay is not ideal. We recommend to keep the default setting.

[7]Here we mean the MRL-E in Kusupati et al. (2022).

Model	Avg.	Pair Class.	M.C.	Structure Prediction		Question Answering			Cross-lingual Retrieval		
		XNLI	XCOPA	UDPOS	WikiANN	XQuAD	MLQA	TyDiQA-GoldP	Mewsli-X	LAReQA	Tatoeba
#Languages (Total 50)		15	11	38	47	11	7	9	38	11	38
Metrics		Acc.	Acc.	F1	F1	F1/EM	F1/EM	F1/EM	mAP@20	mAP@20	Acc.
mBERT-base	59.43	66.63	55.49	71.80	**62.34**	66.23 / 51.03	57.37 / 42.44	**55.01 / 38.05**	**44.65**	**75.26**	39.49
XLM-R-base	62.02	**74.50**	50.45	**73.84**	61.23	72.83 / 58.01	61.54 / 46.45	53.09 / 37.11	42.09	63.43	67.20
mGTE-MLM-2048	**65.24**	73.17	**63.62**	73.25	60.87	**75.33 / 60.00**	64.02 / 48.57	53.58 / 36.68	44.41	72.13	**72.02**
mGTE-MLM-8192	64.44	73.37	61.98	73.14	59.83	74.81 / 59.37	**64.24 / 48.80**	49.85 / 33.27	44.52	71.54	71.10

Table 1: XTREME-R (Ruder et al., 2021) results in the cross-lingual zero-shot transfer (models are trained on English data) setting. M.C. stands for Multiple Choice. The EM scores are not included in the average.

Model	Params	Pos.	Seq. Len.	GLUE Avg.
RoBERTa-base$^{\alpha}$	125M	Abs.	512	86.4
XLM-R-base	279M	Abs.	512	80.44
mGTE-MLM-2048	305M	RoPE	2048	83.42
mGTE-MLM-8192			8192	**83.47**

Table 2: GLUE (Wang et al., 2018) devset averages (w/o WNLI). The detailed scores for each subset are shown in Table 13. $^{\alpha}$Taken from Table 8 of Liu et al. (2019). The rest are from our runs, refer to Appendix C.2.

We fine-tune our contrastive pre-trained embedding model on diverse high-quality datasets with hard-negatives (*e.g.,* MS MARCO (Nguyen et al., 2016), MIRACL (Zhang et al., 2023b), listed in Table 11). We adopt a dynamic batching strategy (Chen et al., 2024) to fine-tune 8192-context data. The batch sampling strategy is the same as the pre-training stage. The τ of MRL and sparse is set to 0.05 and 0.01 respectively. Other details refer to Appendix B.3. We denote this fine-tuned model as `mGTE-TRM`.

2.3 Text Reranking Model

We also build a reranker using the cross-encoder architecture. It takes the query q and document d together as input: [CLS] q [SEP] d, and directly predicts their relevance score by the [CLS] output state: $s_{\text{rerank}} = \boldsymbol{W}\boldsymbol{h}_{[\text{CLS}]}$. In our experiment, $\boldsymbol{W} \in \mathbb{R}^{H\times 1}$ is randomly initialized.

The model is fine-tuned by InfoNCE in one step[8] based on our pre-trained 8k-context text encoder model. Unless otherwise specified, we employ identical data and training settings as our TRM fine-tuning stage (§2.2). The difference lies in our adjustment of the hard-negatives. We describe the detailed settings in Appendix B.4. We denote this model as `mGTE-reranker`.

[8]We found that the contrastive pre-training of reranker does not improve the performance.

Model	Seq.	en	zh	fr	pl
BGE-M3-unsupervised†	8192	56.48	57.53	57.95	55.98
mGTE-CPT	512*	**60.16**	**58.67**	59.72	**57.66**
	8192	60.04	58.63	**59.74**	57.11
mE5-base	514	59.45	56.21	56.19	55.62
mE5-large	514	**61.50**	58.81	56.07	60.08
BGE-M3 (Dense)†	8192	59.84	60.80	58.79	**60.35**
mGTE-TRM (Dense)	8192	61.40	**62.72**	**59.79**	58.22
E5-mistral-7b	32768	66.63	60.81	48.33	-
voyage-multilingual-2	32000	-	-	61.65	-
Cohere-multilingual-v3.0	512	64.01	-	56.02	-
OpenAI-3-large	8191	64.59	-	-	-
OpenAI-3-small	8191	62.26	-	-	-

Table 3: Embedding model performance on MTEB English (Muennighoff et al., 2023a), Chinese (Xiao et al., 2024), French (Ciancone et al., 2024) and Polish (Poświata et al., 2024). The scores of other models are retrieved from the MTEB online leaderboard. *To be consistent with the setting in contrastive pre-training, in retrieval tasks, the max sequence length of the document side is set to 1024. †Denote our runs.

3 Evaluation

We separately evaluate our text encoder in §3.1, TRM and reranker in §3.2 and §3.3.

3.1 Natural Language Understanding

We evaluate the encoder on the cross-lingual natural language understanding (NLU) benchmark XTREME-R[9] (Ruder et al., 2021) and the English NLU benchmark GLUE (Wang et al., 2018). Results show that our encoder outperforms the same-sized previous state-of-the-art XLM-R (Conneau et al., 2020) on all benchmarks.

XTREME-R We focus on the *zero-shot cross-lingual transfer* setting where models are fine-tuned on English trainset and tested on multi- and cross-lingual data. The fine-tuning setup is described in Appendix C.1. We run mBERT-base,

[9]We use XTREME-R (Ruder et al., 2021) instead of XTREME (Hu et al., 2020) since we found the retrieval tasks of XTREME is unstable and difficult to evaluate.

Metric #languages (Total 33)	Params	Seq. Len.	Avg.	MLDR nDCG@10 13	MIRACL nDCG@10 18	MKQA recall@20 25	BEIR nDCG@10 1	LoCo nDCG@10 1
BM25	-	-	47.0	53.6	31.9	28.1	41.7	79.9
mE5-base	279M	514	53.5	30.5	62.3	53.7	48.9	72.2
mE5-large	560M	514	57.7	34.2	65.4	63.5	51.4	74.3
E5-mistral-7b	7111M	32768	62.4	42.6	62.2	62.4	56.9	87.8
OpenAI-3-large	-	8191	-	-	54.9	62.1	55.4	79.4
BGE-M3 Dense			64.3	52.5	67.7	67.8	48.7	84.9
BGE-M3 Sparse	568M	8192	55.1	62.2	53.9	36.3	38.3	84.9
BGE-M3 Dense + Sparse			67.7	64.8	**68.9**	**68.1**	49.4	87.4
mGTE-TRM Dense			66.7	56.6	62.1	65.8	51.1	88.9
mGTE-TRM Sparse	304M	8192	57.2	71.0	55.9	31.6	39.2	88.1
mGTE-TRM Dense + Sparse			**68.9**	**71.3**	64.5	66.0	**51.4**	**91.3**

Table 4: Retrieval results on MIRACL (Zhang et al., 2023b) and MLDR (Chen et al., 2024) (multilingual), MKQA (Longpre et al., 2021) (crosslingual), BEIR (Thakur et al., 2021) and LoCo (Saad-Falcon et al., 2024) (English).

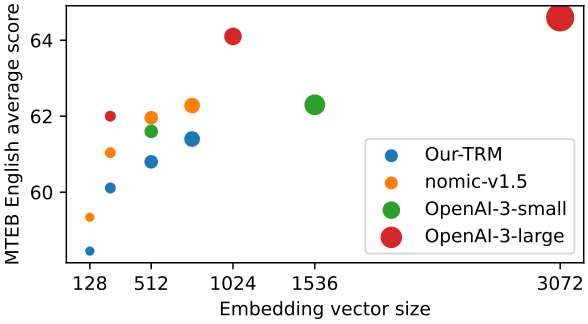

Figure 4: Elastic embedding results on MTEB English.

XLM-R-base, and our encoder, as shown in Table 1. Our 2048 and 8192 encoder models achieve average scores that are higher than those of XLM-R by 3.22 and 2.42 points, respectively.

GLUE We also report the performance on the dev-set of GLUE benchmark (Wang et al., 2018). The fine-tuning details refer to Appendix C.2. Table 2 presents the average scores (Table 13 provides the full results). Our models consistently outperform XLM-R-base and reasonably lag behind the English RoBERTa-base (Liu et al., 2019).

3.2 Text Embedding

Our contrastive pre-training actually yields a text embedding model. To understand the pre-training and fine-tuning of TRM, and to compare with other models, we first run the most popular text embedding benchmark MTEB (Muennighoff et al., 2023a) as well as its Chinese, French and Polish versions.

Multilingual MTEB The results in Table 3 also present the scores of LLM-based models and commercial APIs for reference. For contrastive pre-trained models, our model outperforms `BGE-M3-unsupervised` (Chen et al., 2024) on all four subsets, through our backbone has fewer params than `XLM-R-large`. Comparing with `BGE-M3` and `mE5` (Wang et al., 2024b), our final TRM achieves best scores on Chinese and French, and is competitive on English.

Elastic Embedding We compare our TRM (only elastic embeddings) with open-source model and commercial APIs on MTEB English (Figure 4). Our model presents close scores to the same-sized English-only `nomic-v1.5`, which is promising for a multilingual model. However, it is still behind OpenAI APIs, which is reasonable since they are guessed to be much larger models.

3.3 Text Retrieval

We conduct evaluations to our TRM and reranker on retrieval benchmarks in multilingual (Miracl (Zhang et al., 2023b) and MLDR (Chen et al., 2024)), crosslingual (MKQA (Longpre et al., 2021)) setting, and the commonly used English BEIR (Thakur et al., 2021) and LoCo (Saad-Falcon et al., 2024). Our models are close to the state-of-the-art large models on Miracl, MKQA and BEIR, while achieve better scores on long-context datasets MLDR and LoCo. Details are in Appendix E.

First-Stage Retrieval We compare our TRM to the hybrid model `BGE-M3` (Chen et al., 2024), dense models like `mE5` (Wang et al., 2024b) and `E5-mistral-7b` (Wang et al., 2024a), and BM25. As shown in Table 4, our TRM consistently outperforms `mE5` and OpenAI APIs, better than `BGE-M3` on MLDR, and close to it on the rest parts.

Metric #languages (Total 33)	Params	Seq. Len.	Avg.	MLDR nDCG@10 13	MIRACL nDCG@10 18	MKQA recall@20 25	BEIR nDCG@10 1
Retrieval (mGTE-TRM Dense)	304M	8192	58.9	56.6	62.1	65.8	50.9
jina-reranker-v2-multilingual	278M	8192	59.4	53.2	65.8	**68.8**	49.7
bge-reranker-v2-m3	568M	8192	65.7	66.8	**72.6**	68.7	54.6
mGTE-reranker	304M	8192	**67.4**	**78.7**	68.5	67.2	**55.4**

Table 5: Results of reranking based on the candidates retrieved by our TRM dense model (refer to Table 4).

Model	Attn.	Unpad.	Encoding Time	Search Latency
BGE-M3	eager	×	1800s	20.35ms
	SDPA-MEA	×	744s	
mGTE-TRM	eager	×	695s	15.07ms
	SDPA-MEA	×	298s	
	eager	✓	675s	
	SDPA-MEA	✓	279s	
	MEA	✓	52s	

Table 6: Dense retrieval efficiency. Encoding time is running MLDR-hi corpus (3806 texts with average 4456 tokens after truncating to maximum 8192) on one A100 GPU with FP16. Search latency is measured on a faiss index with 8.8M texts. MEA is the memory-efficient attention in `xFormers`. SDPA-MEA denotes MEA dispatched by scaled dot-product attention of `PyTorch`.

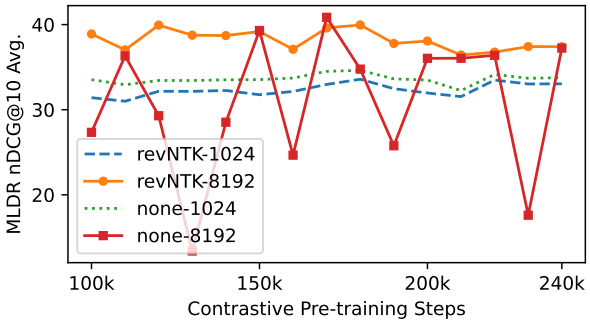

Figure 5: MLDR scores in contrastive pre-training. `none` keeps the RoPE untouched in pre-training. 1024 and 8192 are the max sequence length in evaluations. `revNTK-8912` recovers the 8k context by NTK scaling.

Reranking In Table 5, we evaluate rerankers based on the candidates retrieved by Our-TRM dense model. Our model outperforms the powerful `bge-reranker-v2-m3` (Chen et al., 2024) with a smaller size. Moreover, it greatly surpasses the same-sized `jina-reranker-v2-multilingual`.

3.4 Analysis

Efficiency We compare the efficiency of our TRM with `BGE-M3` on dense retrieval in Table 6. To simulate the real-world scenario, the encoding time is the duration of encoding texts without length grouping. Our TRM is up to 14 times faster than `BGE-M3` (52s *v.s.* 744s). The end-to-end unpadding with `xFormers` is crucial for encoding, which reduces the time by 5 times (52s *v.s.* 279s).

Scaled Contrastive Pre-Training We utilize the reversed NTK scaling in contrastive pre-training to reduce required text length, where we set the RoPE base to $1/8$ of the original and train the 8k encoder with 1k max length. To evaluate the effectiveness, we run the same training without the reversed NTK, comparing the MLDR scores in Figure 5. With revNTK, models exhibit slightly lower performance on 1k context but achieve more stable 8k performance across different training steps.

4 Related Work

Training long-context TRMs has become a hot topic recently. OpenAI released 8191 context APIs (Neelakantan et al., 2022) have set the target for open-source community. Portes et al. (2023) and Günther et al. (2023) replace position embedding of BERT with Alibi (Press et al., 2022) attention bias and pre-train from scratch, which is shown to be effective in build 8k TRMs. Nussbaum et al. (2024) explore the more powerful RoPE (Su et al., 2024) in BERT pre-training and their 2048-context pre-trained encoder achieve better retrieval performance on English. Zhu et al. (2024) suggest patch E5 (Wang et al., 2022) with RoPE. We also use RoPE and provide multi-stage training for native 8192-context text encoder, TRM, and reranker.

Chen et al. (2024) propose long-context multilingual TRM and reranker based on XLM-RoBERTa-large (Conneau et al., 2020) by extending position embedding to 8192 via continue training. We pre-train native 8k multilingual models from scratch for better long-context performance and efficiency.

5 Conclusion

We present the holistic practice of building native 8192-context multilingual retrieval models. We first suggest a text encoder with RoPE and unpadding, which is pre-trained by a two-stage MLM curriculum for 8k context. Evaluations on NLU benchmarks show that our encoder outperforms XLM-RoBERTa in the same size. Based on our encoder, we construct a hybrid TRM and a cross-encoder reranker by contrastive learning. The TRM is pre-trained with reversed RoPE NTK scaling and fine-tuned to generate both Matryoshka embeddings and sparse representations. Results on monolingual and crosslingual retrieval benchmarks show that our TRM and reranker are close to larger ones on regular datasets, and achieve better performance on long-context datasets. This means our models are more efficient for industrial applications.

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Appendix

A MLM Pre-Training

In this section, we describe the data and training configurations of the MLM pre-training of our suggested text encoder.

A.1 Data

Our multilingual pre-training data are composed from following sources:

- C4 (Raffel et al., 2020),
- Skypile (Wei et al., 2023) (2021-2023 subsets),
- mC4 (Xue et al., 2021) (excluded English),
- CulturaX (Nguyen et al., 2024),
- Wikipedia (Foundation),
- books (proprietary).

We filter them and curate a dataset with 1,028B tokens (by XLM-R tokenizer), covering 75 languages (Chinese Simplified and Traditional are counted as one). Table 7 presents the statistics of our final dataset.

A.2 Training Details

We pre-train out text encoder with a two-stage curriculum by masked language model (MLM) objective. The first stage model is trained on maximum length 2048 with batch size 8192 for roughly 0.6 epoch (250k steps) on sampled data (by XLM sampling Eq.1). In the second stage, we down sample texts shorter than 2048 and continue train the model for 30k steps with maximum length 8192 and batch size 2048. The RoPE base is set to $10,000$ and $160,000$ for the first and second stage, respectively (Xiong et al., 2024; Liu et al., 2024; Men et al., 2024).

The text encoder is initialized in base size (12 layers of hidden state size 768) by PyTorch default initialization. We train the model by `transformers` library (Wolf et al., 2020) in BF16 precision. Following Portes et al. (2023), we use the learning rate decoupled AdamW optimizer with weight decay 1e-5. The other hyper-parameters are in Table 8. During training, we split texts that exceed the max sequence length into chunks, but we do not modify shorter texts.

The 250k steps of first stage, MLM-2048, took 10.75 days on 32 A100 80G GPUs. The 30k steps of second stage, MLM-8192, took 20.5 hours on 32 A100 80G GPUs. We acknowledge that this is not the optimal setting and recommend further explorations to optimize the pre-training.

A.3 Additional Discussion on RoPE

We chose RoPE (Su et al., 2024) (to replace absolute position embedding) due to its advantageous properties. RoPE offers excellent context extension capabilities, allowing models to be trained on shorter context windows and then run inference on longer ones. Additionally, it implements asymmetric relative distance encoding, meaning $D(i, j) \neq D(j, i)$, which appears to be particularly important for the training of BERT-like encoder-only models that rely on bidirectional attention. Furthermore, the effectiveness of RoPE has been empirically validated by numerous models, such as RoFormer (Su et al., 2024) and LLaMA (Touvron et al., 2023).

B Contrastive Learning

In this section, we describe the data and training configurations of the contrastive learning of our TRM and reranker.

B.1 Pre-Training Data

Following previous studies, we create large-scale weakly correlated text pairs from diverse sources. The data are primarily consisted of four parts: English pairs (Wang et al., 2022; Li et al., 2023),

ISO code	Language	**Tokens** (M)	**Size** (GiB)	ISO code	Language	**Tokens** (M)	**Size** (GiB)
af	Afrikaans	1,489.19	5.30	ky	Kyrgyz	500.40	3.27
ar	Arabic	14,549.36	79.53	lo	Lao	2.43	0.01
az	Azerbaijani	688.72	3.13	lt	Lithuanian	1,824.46	6.38
be	Belarusian	1,090.61	6.17	lv	Latvian	1,823.43	6.38
bg	Bulgarian	1,454.57	8.94	mk	Macedonian	735.46	4.89
bn	Bengali	1,291.58	9.21	ml	Malayalam	778.66	7.27
ca	Catalan	1,294.05	4.65	mn	Mongolian	958.83	5.91
ceb	Cebuano	633.06	2.02	mr	Marathi	861.05	7.48
cs	Czech	1,465.00	5.27	ms	Malay	96.37	0.39
cy	Welsh	582.49	1.84	my	Burmese	902.46	7.26
da	Danish	1,030.30	4.01	ne	Nepali	657.65	6.32
de	German	18,097.31	67.90	nl	Dutch	5,137.98	18.65
el	Greek	874.87	5.09	no	Norwegian	992.51	3.91
en	English	187,110.31	771.79	pa	Punjabi	726.41	4.96
es	Spanish	148,713.06	601.04	pl	Polish	2,949.88	10.42
et	Estonian	1,111.31	4.10	pt	Portuguese	49,594.59	198.64
eu	Basque	787.46	2.99	qu	Quechua	0.07	0.00
fa	Persian	1,203.16	7.22	ro	Romanian	2,215.05	7.98
fi	Finnish	949.88	3.73	ru	Russian	93,966.28	597.92
fr	French	136,785.00	512.28	si	Sinhala	878.65	7.03
gl	Galician	772.47	3.22	sk	Slovak	884.38	3.31
gu	Gujarati	973.27	6.95	sl	Slovenian	1,100.81	4.05
he	Hebrew	1,842.74	8.36	so	Somali	0.82	0.00
hi	Hindi	1,032.67	8.27	sq	Albanian	700.78	2.73
hr	Croatian	480.19	1.54	sr	Serbian	1,139.38	6.84
ht	Haitian	0.03	0.00	sv	Swedish	840.00	3.37
hu	Hungarian	1,341.23	5.10	sw	Swahili	31.58	0.13
hy	Armenian	805.98	4.88	ta	Tamil	926.84	8.54
id	Indonesian	25,564.33	119.84	te	Telugu	857.91	7.01
is	Icelandic	987.89	3.63	th	Thai	12,782.08	119.52
it	Italian	11,068.23	40.50	tl	Filipino	275.16	1.01
ja	Japanese	135,684.28	601.19	tr	Turkish	1,065.05	4.42
jv	Javanese	0.62	0.00	uk	Ukrainian	893.70	5.68
ka	Georgian	834.90	7.25	ur	Urdu	1,051.83	6.19
kk	Kazakh	1,020.27	6.57	vi	Vietnamese	67,850.87	305.51
km	Khmer	746.15	6.54	yo	Yoruba	0.04	0.00
kn	Kannada	919.83	7.15	zh-cn	Chinese (Simplified)	43,727.30	167.23
ko	Korean	22,865.85	91.78	zh-tw	Chinese (Traditional)	73.39	0.26

Table 7: MLM pre-training data, where we have a total of 1,028B tokens (by XLM-RoBERTa tokenizer). The raw texts are stored in 4.47 TiB arrow files. We report the list of 75 languages (Chinese Simplified and Traditional are counted as one) and include the number of tokens and the size of the data (arrow files, in GiB) for each language.

Hyper-param	MLM-2048	MLM-8192
Number of Params	304M	
Number of Layers	12	
Hidden Size	768	
FFN Inner Size	3072	
Number of Attention Heads	12	
Attention Head Size	64	
Dropout	0.1	
Attention Dropout	0	
Learning Rate Decay	Linear	
Adam ϵ	1e-6	
Adam β_1	0.9	
Adam β_2	0.98	
Gradient Clipping	0.0	
Precision	PyTorch BF16 AMP	
Weight Decay	1e-5	
Max Length	2048	8192
Batch Size	8192	2048
Peak Learning Rate	5e-4	5e-5
Warm-up Ratio	0.06	0.06
Max Steps	250000	30000
RoPE base	10000	160000

Table 8: MLM pre-training hyper-parameters.

Chinese pairs (Li et al., 2023; Xiao et al., 2024), multilingual pairs (cc-news[10]), and crosslingual instruction and translation pairs (Muennighoff et al., 2023b; Team et al., 2024). We filter the data by removing duplicates and low-quality pairs, resulting in a total of 2,938.8M pairs. Table 9 lists the statistics of our contrastive pre-training data (cc-news is separately presented by languages in Table 10).

B.2 Fine-Tuning Data

We collect publicly available high-quality dataset as our fine-tune data as detailed in Table 11. For English, we utilize seven datasets: MS MARCO (Nguyen et al., 2016), Natural Questions (NQ) (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), HotpotQA (Yang et al., 2018), SQuAD (Rajpurkar et al., 2016), FEVER (Thorne et al., 2018), AllNLI from SimCSE (Gao et al., 2021b). For Chinese, we compile six datasets: DuReader (Qiu et al., 2022), mMARCO-zh (Bonifacio et al., 2021), T2-Ranking (Xie et al., 2023a), CmedQAv2 (Zhang et al., 2018), SimCLUE[11], Multi-CPR (Long et al., 2022). Additionally, we incorporate three multilingual datasets: Mr.TyDi (Zhang et al., 2021), MIRACL (Zhang et al., 2023b), and MLDR (Chen et al., 2024). We exclusively use the trainset of each dataset and employ our contrastive pre-trained model to mine hard negatives.

[10] commoncrawl.org/blog/news-dataset-available
[11] https://github.com/CLUEbenchmark/SimCLUE

B.3 TRM Training Setup

Here we separately describe the training setting of the contrastive pre-training and TRM fine-tuning.

Contrastive Pre-Training In the contrastive pre-training, we train a dense representation model (embedder) which take the [CLS] hidden state as the embedding of the input. We use the same XLM sampling strategy (eq.1) to sample batches from each source of Table 9 or cc-news subset of Table 10, where the texts of one batch only come from one single source, and the batch size is $16,384$. We train the model by `transformers` with `deepspeed` ZeRO (Rajbhandari et al., 2020) stage 1 in FP16 precision for roughly 0.4 epoch (240k steps, took 154 hours on 16 A100 80G GPUs) of our data (3.93B pairs on sampled data by Eq.1). We use the AdamW optimizer with the learning rate 2e-4, linear decay, and warm-up ratio 0.05. The $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e-07$. We set gradient clipping to 1.0.

TRM Fine-Tuning In the fine-tuning stage, we further train our embedding model with high-quality datasets as detailed in §B.2. For each query, we incorporate one positive passage and 8 hard negative passages. To enhance long-context retrieval capabilities and maximize training efficiency, we adopt a dynamic batch size strategy as previous work (Chen et al., 2024). Firstly, we group the training data according to their lengths for each dataset. Different batch sizes are then used for varying lengths during training. Additionally, we divide the entire batch into multiple sub-batches, encoding each sub-batch iteratively with gradient checkpointing (Chen et al., 2016) and then gather them to get the final batch's embeddings. We train the embedding model with 10 epochs with 8 A100 80G GPUs. All other hyper-parameters remain consistent with those used in the contrastive pre-training stage. In Table 12, we list the batch size of different length.

B.4 Reranker Training Setup

We utilize the identical fine-tuning dataset for both the reranker and the TRM. For each query, we introduce 10 negative samples, comprising 6 hard negatives and 4 randomly selected negatives. All training parameters expect batch size are kept consistent with those employed for the TRM. The batch sizes are listed in Table 12.

Source	Language	Pairs (M)	Size (GiB)	Source	Language	Pairs (M)	Size (GiB)
agnews	English	1.15	0.30	stackoverflow_title_body	English	18.01	20.49
amazon_qa	English	1.10	0.37	wikihow	English	0.13	0.03
amazon_review_title_body	English	87.86	43.58	wikipedia	English	33.17	19.39
arxiv_title_abstract	English	2.26	2.26	yahoo_body_answer	English	0.68	0.44
baai_mtp_en	English	196.60	178.70	yahoo_qa	English	1.20	0.55
beir_dbpedia	English	4.64	1.59	yahoo_question_body	English	0.66	0.20
beir_debate	English	0.38	0.63	baai_mtp_zh	Chinese	100.13	231.42
beir_pubmed_title_abstract	English	0.13	0.19	baidu_baike	Chinese	34.21	39.05
biorxiv_title_abstract	English	0.20	0.32	baike_qa_train	Chinese	1.43	1.34
clueweb	English	3.94	6.62	commoncrawl_zh	Chinese	28.42	92.79
clueweb_anchor	English	4.51	7.69	gpt3_qa_all	Chinese	4.97	2.39
cnn_dailymail	English	0.31	1.28	gpt3_summarization	Chinese	4.48	1.62
commoncrawl	English	139.94	506.84	medical_quac_wenda_10m	Chinese	10.00	4.55
dpr_reddit	English	199.82	125.71	medical_scholar	Chinese	8.43	7.81
gooaq_qa	English	3.01	0.97	qcl	Chinese	7.40	43.23
hlp_wikipedia	English	19.48	13.55	web_text_zh_train	Chinese	4.12	2.07
medrxiv_title_abstract	English	0.20	0.32	wikipedia	Chinese	4.45	1.07
msmarco	English	2.89	19.56	wodao	Chinese	59.13	190.29
npr	English	0.59	1.03	zh_sft_data_v1	Chinese	0.45	0.43
reddit_title_body	English	124.89	90.36	zh_sft_data_v2	Chinese	2.24	1.37
s2orc_citation_abstract	English	30.58	67.81	zhihu_qa	Chinese	53.42	40.99
s2orc_citation_title	English	51.03	10.84	zhihu_title_body	Chinese	0.94	0.29
s2orc_title_abstract	English	41.77	30.29	xp3x	Crosslingual	351.87	463.85
stackexchange_qa	English	3.00	3.36	translation_eg_NLLB	Crosslingual	940.63	323.06
stackexchange_title_body	English	4.74	4.00				

Table 9: Contrastive pre-training data, where cc-news multilingual data are not included (Table 10). For this Table, we have a total of 2,595.57M pairs (raw texts stored by 2.55 TiB jsonl files).

Lang.	Pairs (M)	Size (GiB)	Lang.	Pairs (M)	Size (GiB)	Lang.	Pairs (M)	Size (GiB)	Lang.	Pairs (M)	Size (GiB)
ar	20.407	32.45	fy	0.044	0.03	lb	0.048	0.05	sk	1.093	1.16
az	0.401	0.23	gl	0.114	0.20	lt	0.321	0.24	sl	1.046	0.93
be	0.039	0.06	gu	0.061	0.06	lv	0.438	0.37	sq	0.282	0.51
bg	3.005	5.03	he	0.397	0.84	mk	0.173	0.44	sr	0.910	1.09
bn	0.463	0.33	hi	14.253	29.90	ml	0.408	0.48	sv	3.361	2.90
ca	0.909	1.30	hr	1.268	1.77	mr	0.278	0.35	sw	0.059	0.07
cs	1.834	2.18	hu	2.668	3.40	my	0.045	0.04	ta	2.125	1.26
da	1.090	1.58	hy	0.125	0.09	nl	6.700	7.41	te	0.355	0.33
de	39.715	57.98	id	6.048	7.46	nn	0.162	0.12	tg	0.038	0.03
el	7.170	14.93	is	0.100	0.05	no	1.978	2.21	th	0.124	0.17
en	0.615	1.47	it	27.827	40.57	or	0.038	0.03	tl	0.055	0.07
es	55.201	86.87	ja	4.139	3.95	pa	0.036	0.04	tr	23.840	26.81
et	0.950	0.85	ka	0.074	0.06	pl	3.530	5.77	uk	5.021	8.42
eu	0.051	0.02	kn	0.192	0.16	pt	12.611	19.28	ur	1.625	0.87
fa	4.839	7.99	ko	8.605	12.48	ro	6.678	9.15	vi	4.375	7.03
fi	1.532	1.93	ky	0.061	0.03	ru	39.451	65.74	MIX*	0.359	0.28
fr	21.242	32.67	la	0.035	0.06	sh	0.220	0.18			

Table 10: The cc-news multilingual pairs (343.26M in total, raw texts stored by 512.8 GiB jsonl files), used in contrastive pre-training together with all data of Table 9. MIX* denotes the mixed pairs of languages that are less than 1GiB (such as af, ceb). We utilize a very large batch size $(16,384)$, and since each batch contains text exclusively from a single source, these low-resource languages might not fill an entire batch. Consequently, we have merged these languages together.

Dataset	Language	Size
MS MARCO, HotpotQA, NQ, NLI, etc.	English	1.4M
DuReader, T^2-Ranking, SimCLUE, etc.	Chinese	2.0M
MIRACL, Mr.TyDi, MLDR	Multilingual	118.9K

Table 11: Specification of training data adopted in Fine-tuning stage.

length	BS(E)	S-BS(R)	BS(E)	S-BS(R)
0-500	768	256	512	256
500-1000	384	128	384	128
1000-2000	256	64	256	64
2000-3000	160	48	160	48
3000-8000	80	16	80	16

Table 12: Batch size (BS) and sub batch size (S-BS) of different length for embedding (E) and reranker (R) model in the fine-tune stage.

C NLU Evaluation

We evaluate our text encoder as well as baselines on the multilingual XTREME-R (Ruder et al., 2021) and English GLUE (Wang et al., 2018) benchmarks. We describe the fine-tuning setup and the evaluation details in the following subsections. The evaluation scripts are available in our github repo[12].

C.1 XTREME-R

We only run XTREME-R (Ruder et al., 2021) in the zero-shot cross-lingual transfer learning setting, where models are fine-tuned on English trainset and tested on multi- and cross-lingual data. We compare our encoder with mBERT-base-cased[13] and XLM-RoBERTa-base[14]. All models are fine-tuned in the same setting and hyper-parameters.

The results are already presented in Table 1.

As XTREME-R has no final release, we implement the evaluation code based on the code of XTREME[15]. However, there are some differences in the retrieval evaluation, where our code will deduplicate the retrieval corpus. In addition, we implement the XCOPA in multiple choice, which might be different from XTREME-R. In fine-tuning, if not specified, we use the epoch number of 3, learning rate of 2e-5, batch size of 32, and max sequence length of 128 (Hu et al., 2020).

[12] github.com/izhx/nlu-evals
[13] hf.co/google-bert/bert-base-multilingual-cased
[14] hf.co/FacebookAI/xlm-roberta-base
[15] github.com/google-research/xtreme

XNLI We fine-tune the model on MNLI[16] (Williams et al., 2018) trainset and then evaluate the checkpoint on XNLI[17] (Conneau et al., 2018).

XCOPA We run this data as the multiple choice task. The model is first trained on SIQA[18] citesap-etal-2019-social and then COPA[19] (Roemmele et al., 2011) for 5 epochs on each dataset. The checkpoint of COPA is evaluated on XCOPA[20] (Ponti et al., 2020).

UDPOS We extract pos-tagging data from the UD (de Marneffe et al., 2021) v2.7 and train the model on trainset of English parts by 10 epochs.

WikiANN We fine-tune the model on the trainset of English by 10 epochs and evaluate on selected WikiANN (Rahimi et al., 2019) testsets[21].

XQuAD We fine-tune on the trainset of SQuAD (Rajpurkar et al., 2016) v1.1[22] for 3 epochs with the learning rate 3e-5 and max length 384. Then we evaluate the checkpoint on XQuAD[23] (Artetxe et al., 2020).

MLQA We directly evaluate the same checkpoint of XQuAD on MLQA[24] (Lewis et al., 2020) with the same setting.

TyDiQA-GoldP We train the model on TyDiQA-GoldP[25] (Clark et al., 2020) trainset in the same setting as XQuAD. Then we evaluate the checkpoint on the testset.

Mewsli-X We generate the data following their github[26]. This is a updated version so that we can not compare with the results in the XTREME-R paper. We train the model on the English wikipedia (mention, entity)-pairs for 2 epochs with the batch size 64 and max length 64. Then we evaluate the checkpoint in the language agnostic retrieval setting, refer to Ruder et al. (2021) for more details.

[16] hf.co/datasets/nyu-mll/glue MNLI subset.
[17] hf.co/datasets/facebook/xnli
[18] hf.co/datasets/allenai/social_i_qa
[19] hf.co/datasets/aps/super_glue copa split.
[20] hf.co/datasets/cambridgeltl/xcopa
[21] hf.co/datasets/unimelb-nlp/wikiann
[22] hf.co/datasets/rajpurkar/squad
[23] hf.co/datasets/google/xquad
[24] hf.co/datasets/facebook/mlqa
[25] hf.co/datasets/juletxara/tydiqa_xtreme
[26] https://github.com/google-research/google-research/blob/master/dense_representations_for_entity_retrieval/mel/mewsli-x.md#getting-started

LAReQA This task is actually conducted on XQuAD-R[27] (Roy et al., 2020). We fine-tune the model on the trainset of SQuAD v1.1 in dual-encoder architecture ([CLS] as the embedding) and retrieval setting for 3 epochs with the batch size 16, max query length 96, and max document length 256. Then we evaluate the checkpoint on XQuAD-R in same setting.

Tatoeba We directly evaluate the checkpoint from LAReQA on Tatoeba[28] (Facebook, 2019) in the same setting.

C.2 GLUE

The GLUE benchmark (Wang et al., 2018) is English transfer learning, *i.e.,* models are trained and tested on the trainset and testset of each dataset (CoLA (Warstadt et al., 2019), SST-2 (Socher et al., 2013), MRPC (Dolan and Brockett, 2005), STS-B (Cer et al., 2017), QQP, MNLI (Williams et al., 2018), QNLI (Rajpurkar et al., 2016), RTE).

We evaluate the GLUE benchmark based on the scripts[29] and data[30] provided by `transformers`. In fine-tuning of each dataset, we use the epoch number of 3, learning rate of 2e-5, batch size of 32, and max sequence length of 128. For MRPC, STS-B, and RTE, we start from the checkpoint of MNLI following (Liu et al., 2019). The MNLI checkpoint is shared with XNLI of XTREME-R (§C.1).

The detailed results are in Table 13. We also include scores of our English models (`Our-en-*`, pre-trained on C4-en) and baselines (Portes et al., 2023; Günther et al., 2023; Nussbaum et al., 2024).

D Text Embedding Evaluation

We have demonstrated the average scores on MTEB English, Chinese, French and Polish (Table 3). In this section, we delve into the details, presenting results of different tasks on each language. For a fair comparison, we do not include the derived models (developed by secondary training on other public off-the-shelf models) in English and Chinese. In addition to the results obtained from the online leaderboard, our own MTEB evaluations were conducted using version 1.2.0 of `mteb` library.

[27] `hf.co/datasets/google-research-datasets/xquad_r`

[28] `hf.co/datasets/mteb/tatoeba-bitext-mining`

[29] `github.com/huggingface/transformers/tree/main/examples/pytorch/text-classification#glue-tasks`

[30] `hf.co/datasets/nyu-mll/glue`

MTEB-en Table 14 shows the results on English MTEB (Muennighoff et al., 2023a). For reference, we include our English embedding models (`Our-en-base/large-embed`, trained by the two-stage contrastive learning on the English part of our data) and top-performing systems from the online leaderboard. We can see that the multilingual models still have a noticeable gap compared to the English models.

MTEB-zh Table 15 presents the C-MTEB (Xiao et al., 2024) (MTEB Chinese subset) results. We include the results of several LLM-based embedding models and APIs. Given that the Chinese community is also keen on optimizing embedding models, the gap between multilingual models and Chinese models is quite noticeable.

MTEB-fr Table 16 demonstrates the F-MTEB (Ciancone et al., 2024) (MTEB French subset) results. Our TRM dense is comparable to the specialized French API `mistral-embed`. However, compared to our our-cpt model, the improvement from fine-tuning is not significant.

MTEB-pl Table 17 lists the Polish MTEB (Poświata et al., 2024) results. Our model does not outperform large-sized BGE and mE5. We speculate this may be due to the limited amount of Polish pairs in the contrastive pre-training, resulting in insufficient training.

E Text Retrieval Evaluation

The retrieval process can be divided into two main stages: recall and reranking. In the recall stage, documents are retrieved using both dense vectors and sparse representations. The final recall score is calculated by weighting the dense retrieval score with a fixed coefficient of 1 and the sparse retrieval score with coefficients ranging from 0.001 to 0.01. Documents not retrieved by either method receive a score of 0. During the ranking stage, the top 100 documents from the recall results are selected as candidates. These candidates are then sorted using our reranker model to produce the final retrieval results.

We present the detail results of MLDR (Chen et al., 2024) (multilingual long-context retrieval, Table 18), MKQA (Longpre et al., 2021) (multilingual, Table 19), MIRACL (Zhang et al., 2023b) (multilingual, Table 20, BEIR (Thakur et al., 2021) (English, Table 21) and LoCo (Saad-Falcon et al., 2024) (English long-context, Table 22).

Model	Params	Pos.	Seq.	Avg.	Single Sentence		Paraphrase and Similarity			Natural Language Inference		
					CoLA	SST-2	MRPC	STS-B	QQP	MNLI	QNLI	RTE
RoBERTa-base$^{\alpha}$	125M	Abs.	512	**86.4**	63.6	94.8	90.2	91.2	91.9	87.6	92.8	78.7
MosaicBERT-base-128$^{\beta}$	137M	Alibi	128	85.4	58.2	93.5	89.0	90.3	92.0	85.6	91.4	83.0
MosaicBERT-base-2048$^{\gamma}$	137M	Alibi	2048	85	54	93	87	90	92	86	92	82
JinaBERT-base$^{\delta}$	137M	Alibi	512	82.6	51.4	94.5	88.4	89.5	80.7	85.7	92.2	78.7
nomic-bert-2048$^{\gamma}$	137M	RoPE	2048	84	50	93	88	90	92	86	92	82
GTEv1.5-en-base-2048	137M	RoPE	2048	85.15	54.46	93.81	93.21	90.00	88.61	86.73	91.67	82.67
GTEv1.5-en-base-8192	137M	RoPE	8192	85.61	57.02	93.35	92.14	90.21	88.78	86.69	91.85	84.84
XLM-R-base	279M	Abs.	512	80.44	30.74	92.43	92.74	89.16	87.74	84.54	90.37	75.81
mGTE-MLM-2048	305M	RoPE	2048	83.42	49.65	92.66	91.17	89.95	88.41	85.40	91.38	78.70
mGTE-MLM-8192	305M	RoPE	8192	**83.47**	48.41	92.32	90.94	89.77	88.50	85.58	91.34	80.87
RoBERTa-large$^{\alpha}$	355M	Abs.	512	**88.9**	68.0	96.4	90.9	92.4	92.2	90.2	94.7	86.6
MosaicBERT-large-128$^{\beta}$	434M	Alibi	128	86.1	59.7	93.7	88.2	90.9	92.0	86.9	93.0	84.5
JinaBERT-large$^{\delta}$	435M	Alibi	512	83.7	59.6	95.0	88.5	88.2	80.9	86.6	92.5	78.5
GTEv1.5-en-large-512	434M	RoPE	512	88.16	64.80	94.50	92.09	91.50	89.23	89.12	93.78	90.25
GTEv1.5-en-large-2048	434M	RoPE	2048	87.02	60.09	94.61	92.14	91.47	89.12	89.02	92.31	87.36
GTEv1.5-en-large-8192	434M	RoPE	8192	87.58	60.39	95.07	93.45	91.37	89.19	89.20	93.90	88.09

Table 13: GLUE (Wang et al., 2018) devset scores (w/o WNLI). $^{\alpha}$Taken from Table 8 of Liu et al. (2019). $^{\beta}$Taken from Table S3 of Portes et al. (2023). $^{\gamma}$Taken from Table 2 of Nussbaum et al. (2024). $^{\delta}$Taken from Table 2 of Günther et al. (2023). The rest of the numbers are from our runs, refer to §C.2 for details.

MTEB English	**Param.**	**Dim.**	**Seq.**	**Avg.**	**Class.**	**Clust.**	**PairC.**	**Rerank.**	**Retr.**	**STS**	**Summ.**
#Datasets (→)				56	12	11	3	4	15	10	1
gte-Qwen2-7b-instruct (Li et al., 2023)	7613M	3584	131072	70.24	86.58	56.92	85.79	61.42	60.25	83.04	31.35
neural-embedding-v1	-	-	-	69.94	87.91	54.32	87.68	61.49	58.12	85.24	30.87
NV-Embed-v1 (Lee et al., 2024a)	7851M	4096	32768	69.32	87.35	52.8	86.91	60.54	59.36	82.84	31.2
voyage-large-2-instruct	-	1024	16000	68.28	81.49	53.35	89.24	60.09	58.28	84.58	30.84
gte-Qwen2-1.5B-instruct (Li et al., 2023)	1776M	1536	131072	67.16	82.47	48.75	87.51	59.98	58.29	82.73	31.17
google-gecko (Lee et al., 2024b)	1200M	768	2048	66.31	81.17	47.48	87.61	58.9	55.7	85.07	32.63
GritLM-7B (Muennighoff et al., 2024)	7242M	4096	32768	66.76	79.46	50.61	87.16	60.49	57.41	83.35	30.37
E5-mistral-7b (Wang et al., 2024a)	7111M	4096	32768	66.63	78.47	50.26	88.34	60.21	56.89	84.63	31.4
text-embedding-3-large	-	3072	8191	64.59	75.45	49.01	85.72	59.16	55.44	81.73	29.92
mxbai-embed-large-v1 (Lee et al., 2024c)	335M	1024	512	64.68	75.64	46.71	87.2	60.11	54.39	85	32.71
nomic-embed-text-v1 (Nussbaum et al., 2024)	137M	768	8192	62.39	74.12	43.91	85.15	55.69	52.81	82.06	30.08
gte-en-large-v1.5	434M	1024	8192	65.39	77.75	47.96	84.53	58.5	57.91	81.43	30.91
gte-en-base-v1.5	137M	768	8192	64.11	77.17	46.82	85.33	57.66	54.09	81.97	31.17
mE5-base (Wang et al., 2024b)	278M	768	514	59.45	73.02	37.89	83.57	54.84	48.88	80.26	30.11
mE5-large (Wang et al., 2024b)	560M	1024	514	61.5	74.81	41.06	84.75	55.86	51.43	81.56	29.69
BGE-m3 (dense)† (Chen et al., 2024)	568M	1024	8192	59.84	74.08	37.27	84.50	55.28	48.82	81.37	31.55
mGTE-TRM (dense)	305M	768	8192	61.40	70.89	44.31	84.23	57.47	51.08	82.11	30.58
BGE-m3-unsupervised† (Chen et al., 2024)	560M	1024	8192	56.48	69.28	38.52	80.92	54.03	42.26	78.30	32.11
mGTE-CPT	305M	768	512*	60.16	72.89	45.05	84.60	58.41	44.93	80.77	29.94
			8192	60.04	72.70	45.35	84.63	58.36	44.46	80.59	30.77

Table 14: Results on MTEB English subset (Muennighoff et al., 2023a). We compare models from the online leaderboard, where derived models (developed by secondary training on other public off-the-shelf models) are not listed. †Denote our runs. *To be consistent with the setting in contrastive pre-training, in retrieval tasks, the max sequence length of the document side is set to 1024.

C-MTEB	Param.	Dim.	Seq.	Avg.	Class.	Clust.	PairC.	Rerank.	Retr.	STS
#Datasets (→)				35	9	4	2	4	8	8
gte-Qwen2-7b-instruct (Li et al., 2023)	7613M	3584	131072	**72.05**	75.09	66.06 8	7.48	68.92	76.03	65.33
piccolo-large-zh-v2 (Huang et al., 2024)	-	-	-	70.95	74.59	62.17	90.24	70	74.36	63.5
OpenSearch-text-hybrid	-	1792	512	68.71	71.74	53.75	88.1	68.27	74.41	62.46
Baichuan-text-embedding	-	1024	512	68.34	72.84	56.88	82.32	69.67	73.12	60.07
gte-Qwen2-1.5B-instruct (Li et al., 2023)	1776M	1536	131072	67.65	71.12	54.61	86.91	68.21	71.86	60.96
E5-mistral-7b (Wang et al., 2024a)	7111M	4096	32768	60.81	70.17	52.3	72.19	61.86	61.75	50.22
mE5-base (Wang et al., 2024b)	278M	768	514	56.21	65.35	40.68	67.07	54.35	61.63	46.49
mE5-large (Wang et al., 2024b)	560M	1024	514	58.81	67.34	48.23	69.89	56	63.66	48.29
BGE-m3 (dense)† (Chen et al., 2024)	568M	1024	8192	60.80	66.95	45.75	73.98	62.88	65.43	52.43
mGTE-TRM (dense)	305M	768	8192	62.72	64.27	47.48	78.34	68.17	71.95	52.73
BGE-m3-unsupervised† (Chen et al., 2024)	560M	1024	8192	57.53	65.04	47.10	64.09	58.14	61.45	48.42
mGTE-CPT	305M	768	512*	58.67	64.64	50.21	63.95	63.77	64.23	46.74
			8192	58.63	64.38	49.84	63.99	64.13	64.30	46.77

Table 15: Results on C-MTEB (Xiao et al., 2024) (MTEB Chinese). We compare models from the online leaderboard, where derived models (developed by secondary training on other public off-the-shelf models) are not listed. †Denote our runs. *To be consistent with the setting in contrastive pre-training, in retrieval tasks, the max sequence length of the document side is set to 1024.

F-MTEB	Param.	Dim.	Seq.	Avg.	Class.	Clust.	PairC.	Rerank.	Retr.	STS	Summ.
#Datasets (→)				26	6	7	2	2	5	3	1
gte-Qwen2-7b-instruct (Li et al., 2023)	7613M	3584	131072	**68.25**	81.76	55.56	90.43	78.7	55.65	82.31	31.45
gte-Qwen2-1.5B-instruct (Li et al., 2023)	1776M	1536	131072	66.6	78.02	55.01	86.88	83.76	52.56	81.26	30.5
voyage-multilingual-2	-	1024	32000	61.65	68.56	46.57	78.66	82.59	54.56	80.13	29.96
voyage-law-2	-	1024	16000	60.58	68.45	44.23	77.3	82.06	52.98	80.29	30.34
mistral-embed	-	1024	-	59.41	68.61	44.74	77.32	80.46	46.81	79.56	31.47
E5-mistral-7b (Wang et al., 2024a)	7111M	4096	32768	48.33	57.72	41.16	76.08	62.2	23.44	65.36	32.22
mE5-base (Wang et al., 2024b)	278M	768	514	56.19	66.8	42.66	74.82	71.76	41.19	77.22	30.76
mE5-large (Wang et al., 2024b)	560M	1024	514	56.07	68.39	38.7	76.19	72.14	42.17	79.37	30.92
BGE-m3 (dense)† (Chen et al., 2024)	568M	1024	8192	58.79	71.57	36.54	79.78	77.36	51.13	80.78	31.05
mGTE-TRM (dense)	305M	768	8192	59.79	68.72	41.66	79.47	76.47	52.97	81.36	29.74
BGE-m3-unsupervised† (Chen et al., 2024)	560M	1024	8192	57.95	69.87	38.43	78.51	75.42	50.05	77.18	28.80
mGTE-CPT	305M	768	512*	59.72	70.79	41.15	80.29	76.19	53.44	76.87	29.04
			8192	59.74	70.69	41.07	79.56	77.10	53.55	77.24	28.74

Table 16: Results on F-MTEB (Ciancone et al., 2024) (MTEB French). We compare top-performing models from the online leaderboard. †Denote our runs. *To be consistent with the setting in contrastive pre-training, in retrieval tasks, the max sequence length of the document side is set to 1024.

MTEB Polish	Param.	Dim.	Seq.	Avg.	Class.	Clust.	PairClass.	Retr.	STS
#Datasets (→)				26	7	1	4	11	3
gte-Qwen2-7b-instruct (Li et al., 2023)	7613M	3584	131072	**67.86**	77.84	51.36	88.48	54.69	70.86
gte-Qwen2-1.5B-instruct (Li et al., 2023)	1776M	1536	131072	64.04	72.29	44.59	84.87	51.88	68.12
mmlw-roberta-large (Dadas et al., 2024)	435M	1024	514	63.23	66.39	31.16	89.13	52.71	70.59
mmlw-e5-large (Dadas et al., 2024)	560M	1024	514	61.17	61.07	30.62	85.9	52.63	69.98
mmlw-roberta-base (Dadas et al., 2024)	124M	768	514	61.05	62.92	33.08	88.14	49.92	70.7
mE5-base (Wang et al., 2024b)	278M	768	514	55.62	59.01	24.97	82.15	44.01	65.13
mE5-large (Wang et al., 2024b)	560M	1024	514	60.08	63.82	33.88	85.5	48.98	66.91
BGE-m3 (dense)† (Chen et al., 2024)	568M	1024	8192	60.35	65.15	25.21	86.46	48.51	69.44
mGTE-TRM (dense)	305M	768	8192	58.22	60.15	33.67	85.45	46.40	68.92
BGE-m3-unsupervised† (Chen et al., 2024)	560M	1024	8192	55.98	60.30	40.17	79.01	43.26	67.05
mGTE-CPT	305M	768	512*	57.66	62.72	38.04	79.70	45.55	67.39
			8192	57.11	61.55	38.15	79.53	45.29	66.53

Table 17: Results on MTEB Polish subset (Poświata et al., 2024) We compare top-performing models from the online leaderboard. †Denote our runs. *To be consistent with the setting in contrastive pre-training, in retrieval tasks, the max sequence length of the document side is set to 1024.

	Max Length	Avg	ar	de	en	es	fr	hi	it	ja	ko	pt	ru	th	zh
BM25	8192	53.6	45.1	52.6	57.0	78.0	75.7	43.7	70.9	36.2	25.7	82.6	61.3	33.6	34.6
$mE5_{large}$	512	34.2	33.0	26.9	33.0	51.1	49.5	21.0	43.1	29.9	27.1	58.7	42.4	15.9	13.2
$mE5_{base}$	512	30.5	29.6	26.3	29.2	45.2	46.7	19.0	40.9	24.9	20.9	50.8	37.8	12.2	12.8
$E5_{mistral\text{-}7b}$	8192	42.6	29.6	40.6	43.3	70.2	60.5	23.2	55.3	41.6	32.7	69.5	52.4	18.2	16.8
BGE-m3-Dense	8192	52.5	47.6	46.1	48.9	74.8	73.8	40.7	62.7	50.9	42.9	74.4	59.5	33.6	26.0
BGE-m3-Sparse	8192	62.2	58.7	53.0	62.1	87.4	82.7	49.6	74.7	53.9	47.9	85.2	72.9	40.3	40.5
BGE-m3-Dense+Sparse	8192	64.8	63.0	56.4	64.2	88.7	84.2	52.3	75.8	58.5	53.1	86.0	75.6	42.9	42.0
mGTE-TRM Dense	8192	56.6	55.0	54.9	51.0	81.2	76.2	45.2	66.7	52.1	46.7	79.1	64.2	35.3	27.4
mGTE-TRM Sparse	8192	71.0	74.3	66.2	66.4	93.6	88.4	61.0	82.2	66.2	64.2	89.9	82.0	47.4	41.8
mGTE-TRM Dense+Sparse	8192	71.3	74.6	66.6	66.5	93.6	88.6	61.6	83.0	66.7	64.6	89.8	82.1	47.7	41.4
+ **mGTE-reranker**	8192	73.8	76.6	70.4	69.3	96.4	89.6	67.8	81.9	68.1	71.1	90.2	86.1	46.7	44.8

Table 18: Evaluation of multilingual long-doc retrieval on the MLDR (Chen et al., 2024) testset (measured by nDCG@10).

	Baselines							M3-Embedding					**mGTE-TRM**			**mGTE-reranker**
	BM25	mDPR	mContriever	$mE5_{large}$	$mE5_{base}$	$E5_{mistral\text{-}7b}$	OpenAI-3	Dense	Sparse	Multi-vec	D+S	All	Dense	Sparse	D+S	ReRank
ar	13.4	33.8	43.8	59.7	44.3	47.6	55.1	61.9	19.5	62.6	61.9	**63.0**	55.9	17.5	56.0	58.2
da	36.2	55.7	63.3	71.7	63.6	**72.3**	67.6	71.2	45.1	71.7	71.3	72.0	69.8	37.9	69.7	71.0
de	23.3	53.2	60.2	**71.2**	62.3	70.8	67.6	69.8	33.2	69.6	70.2	70.4	68.9	27.0	69.1	70.1
es	29.8	55.4	62.3	70.8	63.8	**71.6**	68.0	69.8	40.3	70.3	70.2	70.7	69.6	35.1	70.0	71.0
fi	33.2	42.8	58.7	67.7	53.0	63.6	65.5	67.8	41.2	68.3	68.4	**68.9**	64.2	35.3	64.5	64.9
fr	30.3	56.5	62.6	69.5	61.2	**72.7**	68.2	69.6	43.2	70.1	70.1	70.8	69.8	36.9	70.4	71.0
he	16.1	34.0	50.5	61.4	37.4	32.4	46.3	63.4	24.5	64.4	63.5	**64.6**	55.4	22.0	55.4	56.5
hu	26.1	46.1	57.1	68.0	55.9	**68.3**	64.0	67.1	34.5	67.3	67.7	67.9	64.6	28.8	65.0	66.1
it	31.5	53.8	62.0	71.2	61.6	**71.3**	67.6	69.7	41.5	69.9	69.9	70.3	69.0	36.2	69.2	70.1
ja	14.5	46.3	50.7	63.1	51.7	57.6	64.2	67.0	23.3	67.8	67.1	**67.9**	65.3	19.5	65.2	67.2
km	20.7	20.6	18.7	18.3	28.2	23.3	25.7	58.5	24.4	59.2	58.9	**59.5**	53.6	21.9	53.8	54.7
ko	18.3	36.8	44.9	58.9	40.4	49.4	53.9	61.9	24.3	63.2	62.1	**63.3**	55.9	21.4	56.1	58.9
ms	42.3	53.8	63.7	70.2	62.4	71.1	66.1	71.6	52.5	72.1	71.8	**72.3**	69.9	47.8	70.2	70.9
nl	42.5	56.9	63.9	**73.0**	65.0	74.5	68.8	71.3	52.9	71.8	71.7	72.3	70.7	47.4	70.9	71.5
no	38.5	55.2	63.0	71.1	62.0	70.8	67.0	70.7	47.0	71.4	71.1	**71.6**	69.1	39.7	69.2	70.2
pl	28.7	50.4	60.9	70.5	57.2	**71.5**	66.1	69.4	36.4	70.0	69.9	70.4	68.4	31.4	68.3	69.6
pt	31.8	52.5	61.0	66.8	58.7	**71.6**	67.7	69.3	40.2	70.0	69.8	70.6	69.6	34.9	69.6	70.7
ru	21.8	49.8	57.9	**70.6**	58.7	68.7	65.1	69.4	29.2	70.0	69.4	70.0	68.5	25.8	68.5	69.6
sv	41.1	54.9	62.7	72.0	61.3	**73.3**	67.8	70.5	49.8	71.3	71.5	71.5	69.5	43.3	69.9	70.6
th	28.4	40.9	54.4	69.7	59.7	57.1	55.2	69.6	34.7	70.5	69.8	**70.8**	65.0	30.6	65.2	66.9
tr	33.5	45.5	59.9	67.3	59.2	65.5	64.9	68.2	40.9	69.0	69.1	**69.6**	67.7	36.0	67.7	69.0
vi	33.6	51.3	59.9	68.7	60.0	62.3	63.5	69.6	42.2	70.5	70.2	**70.9**	69.4	37.6	69.3	70.3
zh_cn	19.4	50.1	55.9	44.3	38.3	61.2	62.7	66.4	26.9	66.7	66.6	**67.3**	68.2	23.2	68.4	69.5
zh_hk	23.9	50.2	55.5	46.4	38.3	55.9	61.4	65.8	31.2	66.4	65.9	**66.7**	63.7	27.8	63.8	65.8
zh_tw	22.5	50.6	55.2	45.9	39.0	56.5	61.6	64.8	29.8	65.3	64.9	**65.6**	63.8	26.6	63.9	65.7
Avg	28.1	47.9	56.3	63.5	53.7	62.4	62.1	67.8	36.3	68.4	68.1	**68.8**	65.8	31.6	66.0	67.2

Table 19: Recall@20 on MKQA (Longpre et al., 2021) dataset for cross-lingual retrieval in all 25 languages. The `All` of M3-Embedding denotes the hybrid retrieval result of dense, sparse, and multi-vec scores.

Model	Avg	ar	bn	en	es	fa	fi	fr	hi	id	ja	ko	ru	sw	te	th	zh	de	yo
BM25	31.9	39.5	48.2	26.7	7.7	28.7	45.8	11.5	35.0	29.7	31.2	37.1	25.6	35.1	38.3	49.1	17.5	12.0	56.1
$mE5_{large}$	65.4	76.0	75.9	52.9	52.9	59.0	77.8	54.5	62.0	52.9	70.6	66.5	67.4	74.9	84.6	80.2	56.0	56.4	56.5
$mE5_{base}$	60.13	71.6	70.2	51.2	51.5	57.4	74.4	49.7	58.4	51.1	64.7	62.2	61.5	71.1	75.2	75.2	51.5	43.4	42.3
$E5_{mistral\text{-}7b}$	62.2	73.3	70.3	57.3	52.2	52.1	74.7	55.2	52.1	52.7	66.8	61.8	67.7	68.4	73.9	74.0	54.0	54.0	58.8
OpenAI-3	54.9	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
BGE-M3-Dense	67.8	78.4	80.0	56.9	55.5	57.7	78.6	57.8	59.3	56.0	72.8	69.9	70.1	78.6	86.2	82.6	61.7	56.8	60.7
BGE-M3-Sparse	53.9	67.1	68.7	43.7	38.8	45.2	65.3	35.5	48.2	48.9	56.3	61.5	44.5	57.9	79.0	70.9	36.3	32.2	70.0
BGE-M3-Multi-vec	69.0	79.6	81.1	59.4	57.2	58.8	80.1	59.0	61.4	58.2	74.5	71.2	71.2	79.0	87.9	83.0	62.7	57.9	60.4
BGE-M3-Dense+Sparse	68.9	79.6	80.7	58.8	57.5	59.2	79.7	57.6	62.8	58.3	73.9	71.3	69.8	78.5	87.2	83.1	62.5	57.6	**61.8**
BGE-M3 All	**70.0**	**80.2**	**81.5**	**59.8**	**59.2**	**60.3**	**80.4**	**60.7**	**63.2**	**59.1**	**75.2**	**72.2**	**71.7**	**79.6**	**88.2**	**83.8**	**63.9**	**59.8**	61.5
mGTE-TRM Dense	62.1	71.4	72.7	54.1	51.4	51.2	73.5	53.9	51.6	50.3	65.8	62.7	63.2	69.9	83.0	74.0	60.8	49.7	58.3
mGTE-TRM Sparse	55.9	66.5	70.4	35.6	46.2	40.0	47.6	66.5	39.8	48.9	47.9	59.3	64.3	47.1	59.4	83.0	70.5	73.7	39.9
mGTE-TRM Dense+Sparse	63.5	73.4	75.1	49.9	57.6	62.7	52.0	74.7	53.5	56.4	52.8	67.1	66.7	63.5	69.5	85.2	75.8	58.4	58.8
+ **mGTE-reranker**	68.5	77.1	63.1	78.6	56.3	72.4	80.3	79.6	58.6	59.1	74.6	75.5	59.4	56.3	56.5	62.2	72.2	86.3	65.1

Table 20: Multi-lingual retrieval performance on the MIRACL (Zhang et al., 2023b) dev set (measured by nDCG@10).

BEIR	Avg.	Argu-Ana	Cli-mate-Fever	CQA-Dup-Stack	DB-Pedia	Fever	FiQA	Hotpot-QA	MS MAR-CO	NF-Corpus	NQ	Quora	Sci-docs	Sci-fact	Touche-2020	Trec-Covid
gte-Qwen2-7B-instruct	60.25	64.27	45.88	46.43	52.42	95.11	62.03	73.08	45.98	40.6	67	90.09	28.91	79.06	30.57	82.26
NV-Embed-v1	59.36	68.2	34.72	50.51	48.29	87.77	63.1	79.92	46.49	38.04	71.22	89.21	20.19	78.43	28.38	85.88
gte-Qwen2-1.5B-instruct	58.29	69.72	42.91	44.76	48.69	91.57	54.7	68.95	43.36	39.34	64	89.64	24.98	78.44	27.89	85.38
voyage-large-2-instruct	58.28	64.06	32.65	46.6	46.03	91.47	59.76	70.86	40.6	40.32	65.92	87.4	24.32	79.99	39.16	85.07
neural-embedding-v1	58.12	67.21	32.3	49.11	48.05	89.46	58.94	78.87	42	42.6	68.36	89.02	27.69	78.82	24.06	75.33
GritLM-7B	57.41	63.24	30.91	49.42	46.6	82.74	59.95	79.4	41.96	40.89	70.3	89.47	24.41	79.17	27.93	74.8
e5-mistral-7b-instruct	56.89	61.88	38.35	42.97	48.89	87.84	56.59	75.72	43.06	38.62	63.53	89.61	16.3	76.41	26.39	87.25
google-gecko	55.7	62.18	33.21	48.89	47.12	86.96	59.24	71.33	32.58	40.33	61.28	88.18	20.34	75.42	25.86	82.62
text-embedding-3-large	55.44	58.05	30.27	47.54	44.76	87.94	55	71.58	40.24	42.07	61.27	89.05	23.11	77.77	23.35	79.56
gte-en-large-v1.5	57.91	72.11	48.36	42.16	46.3	93.81	63.23	68.18	42.93	36.95	56.08	89.67	26.35	82.43	22.55	77.49
gte-en-base-v1.5	54.09	63.49	40.36	39.52	39.9	94.81	48.65	67.75	42.62	35.88	52.96	88.42	21.92	76.77	25.22	73.13
BM25	41.7	31.5	21.3	29.9	31.3	75.3	23.6	60.3	22.8	32.5	32.9	78.9	15.8	66.5	36.7	65.6
mE5-large	51.43	54.38	25.73	39.68	41.29	82.81	43.8	71.23	43.7	33.99	64.06	88.18	17.47	70.41	23.39	71.33
mE5-base	48.88	44.23	23.86	38.52	40.36	79.44	38.17	68.56	42.27	32.46	60.02	87.65	17.16	69.35	21.35	69.76
BGE-M3 Dense†	48.34	53.95	29.52	39.09	39.80	81.38	41.30	69.44	38.32	31.43	60.60	88.57	16.39	64.36	22.63	55.59
BGE-M3 Sparse†	38.30	25.08	24.69	27.51	23.21	88.36	26.79	68.45	19.59	27.5	17.98	73.82	8.89	64.37	30.26	48.00
BGE-M3 Dense+Sparse†	49.41	53.88	30.21	39.10	39.89	81.24	40.25	70.11	37.62	32.53	59.58	88.62	15.59	65.74	31.12	55.67
mGTE-TRM Dense	51.07	58.36	34.83	38.12	40.11	92.07	44.99	63.03	39.92	36.66	58.10	88.02	18.26	73.42	22.76	57.4
mGTE-TRM Sparse	39.24	40.06	24.17	25.11	20.0	88.32	28.58	64.68	19.39	28.34	19.71	76.84	10.92	67.72	21.52	53.33
mGTE-TRM Dense+Sparse	51.43	58.48	34.89	38.36	39.72	93.14	44.98	65.01	39.99	36.67	56.90	89.05	18.26	73.45	24.09	58.46
+ **mGTE-reranker**	55.42	58.53	44.93	38.37	45.62	93.9	44.38	74.51	44.99	36.29	65.21	81.67	18.42	75.59	31.29	77.75
BGE-M3-unsupervised†	42.26	59.07	23.05	38.10	31.16	59.15	36.57	53.39	27.79	30.67	39.69	86.38	15.08	61.26	17.62	54.90
mGTE-CPT-512,1024	44.93	52.99	17.93	45.01	37.63	34.13	48.38	54.39	31.76	39.01	48.48	86.82	22.95	72.46	18.56	63.46
mGTE-CPT-8192	44.46	55.14	15.85	44.73	38.74	27.42	47.45	55.93	31.79	38.62	49.27	86.81	22.72	73.08	17.08	62.27

Table 21: BEIR benchmark (Thakur et al., 2021) nDCG@10 scores. We include top models from MTEB Retrieval English leaderboard. †Denote our runs.

Model	**Param.**	**Dim.**	**Seq**	**Avg.**	Tau Scr.	Tau Gov.	Tau QMS.	QASP. Tit. Art.	QASP. Abs. Art.
$\text{Jina}_{\texttt{base-v2}}$ (Günther et al., 2023)	137M	768	8192	85.5	93.3	98.6	40.8	95.1	99.3
nomic-embed-text-v1 (Nussbaum et al., 2024)	137M	768	8192	85.5	90.9	97.8	44.2	94.9	99.9
text-embedding-3-small	-	1536	8192	82.4	92.2	97.7	27.4	95.9	98.9
text-embedding-3-large	-	3072	8192	79.4	88.0	93.6	25.5	93.2	96.8
mGTE-en-base-embed	137M	768	8192	87.4	91.8	98.6	49.9	97.1	99.8
mGTE-en-large-embed	434M	1024	8192	86.7	92.6	98.7	44.5	97.8	99.8
mE5_{base} (Wang et al., 2024b)	279M	768	512	72.2	68.9	87.6	30.5	85.1	88.9
$\text{mE5}_{\text{large}}$ (Wang et al., 2024b)	279M	1024	512	74.3	70.4	89.5	37.6	89.5	85.4
$\text{E5}_{\texttt{mistral}}$ (Wang et al., 2024a)	7B	4096	4096	87.8	95.9	98.3	46.8	98.4	99.8
BGE-M3-Dense† (Chen et al., 2024)	568M	1024	8192	84.9	93.8	97.4	41.9	93.2	98.3
BGE-M3-Sparse† (Chen et al., 2024)	568M	1024	8192	84.9	95.5	97.9	46.7	85.7	98.9
BGE-M3-Dense+Sparse† (Chen et al., 2024)	568M	1024	8192	87.4	97.7	98.2	47.7	93.6	99.7
mGTE-TRM Dense	434M	768	8192	88.9	95.1	97.7	58.5	94.6	98.7
mGTE-TRM Sparse	434M	768	8192	88.1	97.6	97.9	60.1	85.5	99.2
mGTE-TRM Dense+Sparse	434M	768	8192	91.3	98.2	98.3	66.5	94.6	98.7

Table 22: The nCDG@10 scores on the LoCo benchmark (Saad-Falcon et al., 2024). †Denote our runs.